

Pushpanath M R

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Professional summary

Results-driven full-stack developer with hands-on experience building web and mobile applications using ReactJS and NextJS. Proficient in REST APIs, Firebase authentication, cloud-native workflows, and Agile development practices. Proven ability to deliver maintainable, user-focused software with measurable impact, backed by a published patent and Google Cloud certifications.

Experience

Full-Stack Developer Intern, CodeStreams Innovations Pvt Ltd – Bangalore, India Aug 2023 – Oct 2023

- Engineered 5+ responsive web interfaces using ReactJS, NextJS, HTML, Tailwind CSS, and JavaScript, improving page layout consistency and usability across 3 device breakpoints.
- Configured secure user authentication flows and real-time database features using Firebase, supporting reliable data handling for 100+ user sessions with zero reported auth failures.
- Validated 10+ REST API endpoints using Postman, identifying and resolving 4 critical edge-case errors before production release.
- Integrated Google Cloud services for authentication and storage, reducing manual setup time by 30% through reusable cloud workflow templates.

5G L1 (PHY Layer) Intern, Nokia Solutions & Networks – [Bangalore, India] [Aug 2025 – Present]

- Engineered support for multi-PUCCH request handling per slot in 5G NR PHY layer, contributing to performance-critical L1 feature development and validation workflows.
- Designed and implemented enhancements to the System Component Test (SCT) framework, including scalable test scenario handling and MATLAB-based test generation for end-to-end validation.
- Optimized SCT test suite efficiency by reducing test file sizes and eliminating redundant payload dumps, improving execution speed and maintainability of validation pipelines.
- Developed strong familiarity with Nokia's internal L1 validation framework, gaining end-to-end exposure from feature specification to system-level testing, while independently exploring Generative and Agentic AI applications.

Education

Vellore Institute of Technology, Chennai, MTech Integrated Software Engineering Sept 2021 – Present
CGPA: 8.07/10 (equiv. 3.2/4.0)

Sri Ayyan Vidyashram Higher Secondary School, R K Pet. Sept 2019 – July 2021
HSC Percentage: 91.5%

Skills

Programming Languages: Java, Python, C/C++, JavaScript

Frontend: ReactJS, NextJS, HTML, CSS, Tailwind CSS

Backend & APIs: NodeJS, ExpressJS, REST APIs, JWT Authentication

Mobile Development: React Native, Expo

Databases: Firebase, MySQL, Oracle SQL, SQLite

Tools: Git, GitHub, Postman, Figma

Cloud Platforms: Google Cloud Platform, Firebase

Core Concepts: Data Structures, Object-Oriented Programming, Responsive Design

Patents

Smart Backpack with Integrated Location Tracking, Power Support, and Emergency Features

Published – 2024 (K&K Ref No: PS00020319)

Developed an innovative IoT-enabled smart backpack featuring real-time location tracking, moisture detection sensors, and an emergency SOS alert system. The design integrates multiple power sources, including piezoelectric energy and solar panels, ensuring sustainable power solutions while enhancing user safety and convenience.

Projects

Iron Arc – Offline-First Fitness Tracking Platform

TechStack : React Native, Expo, TypeScript, SQLite, Firebase Auth, Firestore

Built a cross-platform offline-first fitness tracking application focused on strength logging and long-term progress analytics. Designed local-first data storage using SQLite with a deferred sync mechanism to batch cloud updates, ensuring fast performance and reliable data handling without continuous connectivity. Structured the application into modular layers covering UI, database, sync logic, and analytics.

Trip360 – Travel Advisor Website

TechStack : Next.js, React.js, Gemini API, Firebase, Google Auth

Developed a travel advisory website focused on attractions across Tamil Nadu. Integrated listings for hotels, restaurants, and flights with real-time distance calculation. Implemented an AI-powered chatbot using Gemini API to provide contextual travel assistance and personalized recommendations.

AI-Driven Smart Backpack for Visually Impaired Individuals

TechStack : YOLOv8, ESP32, GPS, Sensors, Python, Embedded C, Firebase, Mobile App

Developed an assistive smart backpack tailored for visually challenged users. Integrated YOLOv8 object detection to identify potholes, vehicles, and obstacles in real time. Implemented audio alerts, live location tracking, moisture detection, emergency SOS signaling, and energy harvesting via solar and piezo modules. Enhanced night-time navigation using LED feedback and voice guidance through a mobile app interface.

Customer Churn Analysis

TechStack : Python, Machine Learning, Pandas, Scikit-Learn

Built a machine learning pipeline to predict customer churn using a telecom dataset with over 7,000 users. Performed data preprocessing, feature engineering, and model evaluation using Random Forest and XGBoost, achieving 80.8% accuracy. Visualized insights through Plotly and Matplotlib to support data-driven decision making.

Certifications

Google UX Design Professional Certificate

Google - 2023

Link: [Google-Ux-Design-Certification](#)

Preparing for Google Cloud Certification: Cloud DevOps Engineer

Google - Coursera - 2022

Link: [Cloud DevOps Engineer](#)

Google Cloud Fundamentals: Core Infrastructure

Google - Coursera - 2023

Link: [Google Cloud Fundamentals: Core Infrastructure](#)

Achievements and Awards

Google Cloud Arcade - Cloud Skills Boost (Season 1 & 2)

Dec 2023 – Jul 2024

- **Season 1 (Dec 2023):** Scored **56,765 points**, secured a position in the **Gold League**, earned **37 skill badges**, and completed multiple hands-on cloud labs and challenges.
- **Season 2 (Jan – Jul 2024):** Scored **74,285 points**, reached the **Diamond League**, earned **33 skill badges**, and enhanced cloud expertise through advanced labs and problem-solving activities.